**King Fahd University of Petroleum and Minerals**
**Department of Mathematics**
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# Math 102 Major Exam II Term 232

Original exam date: not printed. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

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**1.** $\int_0^3 xe^{x/2}\,dx =$

(a) $3e^{3/2} - 1$

(b) $4 + 2e^{3/2}$

(c) $3 + 2e^{3/2}$

(d) $2e^{3/2} - 1$

(e) $3e^{3/2}$

**2.** $\int \frac{\tan^2 x}{\sec^5 x}\,dx =$

(a) $\frac{1}{3}\tan^3 x - \frac{1}{7}\tan^6 x + c$

(b) $\frac{1}{\sqrt{3}}\sin^3 x - \frac{1}{6}\sin^6 x + c$

(c) $\frac{1}{5}\cos^3 x - \frac{1}{3}\sin^5 x + c$

(d) $\frac{1}{\sqrt{3}}\cos^3 x - \frac{1}{5}\sin^5 x + c$

(e) $\frac{1}{3}\sin^3 x - \frac{1}{5}\sin^5 x + c$

**3.** $\int \sin^4 x\,dx =$

(a) $x - \frac{1}{2}\sin 2x - \frac{1}{16}\sin 4x + c$

(b) $\frac{1}{4}x + \frac{1}{4}\sin 2x + \frac{1}{8}\sin 4x + c$

(c) $4x - \frac{3}{4}\sin 2x - \frac{1}{16}\sin 4x + c$

(d) $\frac{5}{8}x + \frac{1}{8}\sin 2x - \frac{1}{16}\sin 4x + c$

(e) $\frac{3}{8}x - \frac{1}{4}\sin 2x + \frac{1}{32}\sin 4x + c$

**4.** $\int \frac{dx}{(x^2+5)^{3/2}} =$

(a) $\dfrac{5x}{\sqrt{x^2+25}} + c$

(b) $\dfrac{1}{5x\sqrt{x^2+5}} + c$

(c) $\dfrac{x}{5\sqrt{x^2+5}} + c$

(d) $\dfrac{\sqrt{5}}{5}\, x\sqrt{x^2+5} + c$

(e) $\dfrac{1}{5}\, x\sqrt{x^2+25} + c$

**5.** The arc length of the graph of $f(x) = \ln(\sin x)$ over $\left[\dfrac{\pi}{4}, \dfrac{3\pi}{4}\right]$ is equal to

(a) $\ln(\sqrt{2}-1)$

(b) $\ln(\sqrt{2}+1)$

(c) $\ln\left(\dfrac{\pi\sqrt{2}-1}{2}\right)$

(d) $\ln\left(\dfrac{\sqrt{2}+1}{\sqrt{2}-1}\right)$

(e) $\ln\left(\dfrac{\pi\sqrt{2}+1}{2}\right)$

**6.** $\int \frac{x^3}{\sqrt{16-x^2}}\, dx =$

(a) $\dfrac{1}{64}\sqrt{16-x^2}\,(x^3+16) + c$

(b) $-16\sqrt{16-x^2} + x\sqrt{(16-x^2)^3} + c$

(c) $4\sqrt{16-x^2} + (16-x^2)^{5/2} + c$

(d) $-\dfrac{1}{3}\sqrt{16-x^2}\,(32+x^2) + c$

(e) $-4\sqrt{16-x^2}\,(16-x^2) + c$

**7.** $\int_e^{e^2} \ln x \, dx =$

(a) $e^2 - e$

(b) $\dfrac{e^2 + e}{2}$

(c) $\dfrac{e^2 - e}{2}$

(d) $e^2 + e$

(e) $e^2$

**8.** Suppose that $f(a) = f(b) = 0$ and the second derivative of $f$ exists on the interval $[a, b]$. Then the value of $\int_a^b (x - a)(x - b) f''(x) \, dx =$

(a) $(b - a) \int_a^b f(x) \, dx$

(b) $(b - a) \int_a^b f(-x) \, dx$

(c) $\dfrac{b + a}{2} \int_a^b f(-x) \, dx$

(d) $2 \int_a^b f(x) \, dx$

(e) $\dfrac{ab}{2} \int_a^b f(x) \, dx$

**9.** $\int \frac{9 - x^2}{7x^3 + x} \, dx =$

(a) $9 \ln |x| - \dfrac{32}{7} \ln(1 + 7x^2) + c$

(b) $7 \ln(1 + 7x^2) + 9x + c$

(c) $7 \ln |3 - x| + 9 \ln |3 + x| + \ln |x| + c$

(d) $\dfrac{7}{9} \ln(7x^3 + x) - x + c$

(e) $18 \ln |x| - \dfrac{9}{7} \ln(1 + 7x^2) + c$

**10.** If the integral $\int \tan^5 \frac{x}{2}\, dx$ is written in the form

$$\int \tan^5 \frac{x}{2}\, dx = a \tan^4 \frac{x}{2} + b \tan^2 \frac{x}{2} + c \ln \left|\cos \frac{x}{2}\right| + C,$$

then the value of $a + b + c =$

(a) $-\frac{7}{2}$

(b) $-\frac{3}{2}$

(c) $-\frac{5}{2}$

(d) 0

(e) $-\frac{1}{2}$

**11.** The volume of the solid generated by revolving about the $y$-axis the region bounded by the graphs $y = x^{3/2}$, $y = 8$, $x = 0$ is equal to:

(a) $\frac{199}{7}\pi$

(b) $\frac{384}{7}\pi$

(c) $\frac{200}{7}\pi$

(d) $\frac{196}{7}\pi$

(e) $\frac{219}{7}\pi$

**12.** The volume of the solid generated by revolving the region bounded by the graphs $y = \sqrt{x}$, $y = -\frac{1}{2}x + 4$, $x = 0$, $x = 8$ about the $x$-axis is equal to:

(a) $56\pi$

(b) $48\pi$

(c) $52\pi$

(d) $42\pi$

(e) $44\pi$

**13.** $\int \frac{e}{e^{-x}+1}\,dx =$

(a) $\ln(e^{-x}+x)+c$

(b) $e\ln(1+e^{x})+c$

(c) $\dfrac{1}{e}\ln(1+e^{-x})+c$

(d) $e^{x}\ln(1+e^{-x})+c$

(e) $\ln(e^{-x}+e)+c$

**14.** $\int \cot^2 x\,dx =$

(a) $-\cot x+\csc x+c$

(b) $\dfrac{\cot^3 x}{3}+c$

(c) $\csc x-2x+c$

(d) $-\csc x-2x+c$

(e) $-\cot x-x+c$

**15.** The surface area that is generated when $f(x)=3x$ over $[0,3]$ is revolved about the $x$-axis is equal to:

(a) $29\pi\sqrt{10}$

(b) $10\pi\sqrt{10}$

(c) $36\pi\sqrt{10}$

(d) $9\pi\sqrt{10}$

(e) $27\pi\sqrt{10}$

**16.** $\int x^3\sqrt{x^4+1}\,dx =$

(a) $\dfrac{1}{6}(x^4+1)^{3/2}+c$

(b) $\dfrac{1}{12}(x^4+1)^{3/2}+c$

(c) $\dfrac{3}{10}(x^4+1)^{3/2}+c$

(d) $\dfrac{2}{3}(x^4+1)^{3/2}+c$

(e) $\dfrac{3}{8}(x^4+1)^{3/2}+c$

**17.** $\int_0^1 2\sin^{-1} x\,dx =$

(a) $\pi - 2$

(b) $2\sin^{-1}(1)$

(c) $\pi + 1$

(d) $\dfrac{\pi}{2} - 1$

(e) $\dfrac{\sin^{-1}(1)}{2}$

**18.** $\int_0^{\pi/2} \frac{\pi}{1+\sin\theta+\cos\theta}\,d\theta =$

(a) $\ln 2$

(b) $\dfrac{1}{2}\ln\left(\dfrac{\pi^2}{2}\right)$

(c) $\dfrac{1}{e}\ln(2\pi)$

(d) $\pi \ln 2$

(e) $\dfrac{1}{\pi}\ln\left(\dfrac{\pi^3}{4}\right)$

# Answer Key

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| Q | Ans | Brief explanation |
|---|---|---|
| 1 | **(b)** | Parts: $\left[2xe^{x/2} - 4e^{x/2}\right]_0^3 = 4 + 2e^{3/2}$. |
| 2 | **(e)** | Integrand $= \sin^2 x \cos^3 x$; $u = \sin x$ gives $\frac{1}{3}\sin^3 x - \frac{1}{5}\sin^5 x$. |
| 3 | **(e)** | $\sin^4 x = \frac{3}{8} - \frac{1}{2}\cos 2x + \frac{1}{8}\cos 4x$, integrate term by term. |
| 4 | **(c)** | $x = \sqrt{5}\tan\theta$: $\frac{1}{5}\int \cos\theta\, d\theta = \frac{x}{5\sqrt{x^2+5}} + c$. |
| 5 | **(d)** | $\sqrt{1+\cot^2 x} = \csc x$, $L = [\ln\lvert\csc x - \cot x\rvert] = \ln\frac{\sqrt{2}+1}{\sqrt{2}-1}$. |
| 6 | **(d)** | $u = 16 - x^2$: $-16\sqrt{u} + \frac{1}{3}u^{3/2} = -\frac{1}{3}\sqrt{16-x^2}(32 + x^2)$. |
| 7 | **(e)** | $[x\ln x - x]_e^{e^2} = e^2 - 0 = e^2$. |
| 8 | **(d)** | Parts twice with $f(a) = f(b) = 0$: the integral equals $2\int_a^b f(x)\,dx$. |
| 9 | **(a)** | $A = 9,\ B = -64,\ C = 0$: $9\ln\lvert x\rvert - \frac{32}{7}\ln(1+7x^2) + c$. |
| 10 | **(c)** | $2\int \tan^5 u\, du = \frac{1}{2}\tan^4 u - \tan^2 u - 2\ln\lvert\cos u\rvert$: $a + b + c = -\frac{5}{2}$. |
| 11 | **(b)** | Shells: $V = 2\pi\int_0^4 x(8 - x^{3/2})dx = 2\pi\cdot\frac{192}{7} = \frac{384}{7}\pi$. |
| 12 | **(b)** | Split at $x = 4$: $V = \pi\left(\frac{88}{3} + \frac{56}{3}\right) = 48\pi$. |
| 13 | **(b)** | Multiply by $e^x/e^x$, $u = 1 + e^x$: result $e\ln(1+e^x) + c$. |
| 14 | **(e)** | $\cot^2 x = \csc^2 x - 1$, so the integral is $-\cot x - x + c$. |
| 15 | **(e)** | $S = 2\pi\int_0^3 3x\sqrt{10}\,dx = 27\pi\sqrt{10}$. |
| 16 | **(a)** | $u = x^4 + 1$: $\frac{1}{4}\int u^{1/2}du = \frac{1}{6}(x^4+1)^{3/2} + c$. |
| 17 | **(a)** | Parts: $2x\sin^{-1}x + 2\sqrt{1-x^2}\Big\rvert_0^1 = \pi - 2$. |
| 18 | **(d)** | $t = \tan\frac{\theta}{2}$: $\pi\int_0^1 \frac{dt}{1+t} = \pi\ln 2$. |